Abdelrahman Khalifa

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Summary

Software engineer who likes building tools and automation for other engineers. Set up multi-architecture CI that tests against real services, built REST backends and firmware-update automation, and writes Python, Rust, C and SQL on Linux. Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting) with 11 PRs merged upstream.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built CI in GitHub Actions on x86_64 and aarch64 runners: builds libcups3 from source, starts real CUPS and Avahi services and runs linting and tests against them; added a CI matrix across architectures and CUPS 2/3 to cups-rs.
- Got **11 PRs merged upstream** through maintainer code review (list), including crash fixes in libcups (C).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Built a Node.js/Express + MongoDB REST API for remote operation, and automated firmware delivery: new builds uploaded in a web app are downloaded by the desktop app and flashed to devices with esptool; GitHub Actions CI.
- Developed the Python (PyQt5) control app; used asyncio to wait for device and server messages concurrently, cutting CPU use and latency by about 40% (measured).

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* Mar 2025 – Dec 2025

- Rust Clippy: fixed `ref_as_ptr` false positives in a lint used by Rust developers everywhere (PR #16214); Boa: faster `JsStr::code_points()` (PR #4553); Zephyr RTOS: `IP_MULTICAST_LOOP` socket option (#86977).

Projects

Developer environment automation | *Bash* GitHub

- Idempotent install script that sets up a full Linux dev environment (packages, configs, editor plugins and language servers); backs up existing files and can be re-run safely, with flags to skip slow steps.

Function Solver and Plotter | *Python, PySide2, SymPy, Matplotlib* GitHub

- Input validation, error handling and end-to-end tests with pytest-qt.

Database Systems, CMU 15-445 | *C++17, SQL* GitHub

- Thread-safe Count-Min Sketch for the BusTub database; SQL analytics in DuckDB (joins, aggregation, upserts).

Technical Skills

Languages: Python (asyncio), SQL, Bash, Rust, C, C++, JavaScript (Node.js)
CI/CD & testing: GitHub Actions (matrix and multi-architecture builds, service setup in CI), automated testing (pytest, Cargo tests, integration tests against real services), linting and formatting checks
APIs & data: REST API design, Express, MongoDB, SQL Server, DuckDB; IPC (varlink)
Engineering practice: Git/GitHub, code review, Linux (Ubuntu, Arch), debugging, technical writing

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Codeforces Expert; 4-time ECPC finalist, 2021–2024; mentor in the local ICPC community (20+ colleagues).